



## Learning Python - Sheet #12

### Spacewar #3

For now, it's not clear that our spaceship is moving except when we press the arrows.

To give the illusion of movement, we'll add a starfield: some stars that move to the left in the background. This will give us the impression that the spaceship is moving towards the right.

At the very top, add this:

```
import random
```

Then also add a new constant:

```
NB_STARS = 200
```

Now add the following functions to your code. These are functions to handle creating and moving the stars.

```
def init_stars():
    stars = []
    for i in range(NB_STARS):
        stars.append((random.randint(0, screen.get_width()), random.randint(0,
screen.get_height())))
    return stars

def move_stars(stars):
    for i in range(NB_STARS):
        stars[i] = (stars[i][0]-1, stars[i][1])
        if stars[i][0] == 0:
            stars[i] = (screen.get_width(), random.randint(0, screen.get_height()))

def draw_stars(screen, stars):
    for i in range(NB_STARS):
        screen.set_at(stars[i], "white")
```

Finally, change the function draw and your game loop (add the calls to the new functions and draw the missiles):

```
def draw():
    screen.fill("black")
    draw_stars(screen, stars)
    screen.blit(spaceship, (x, y))
    for m in missiles:
        screen.blit(missile, m)
    pygame.display.flip()

stars = init_stars()

while running:
    handle_events()
    move_spaceship()
    fire_missiles()
    move_missiles()
    move_stars(stars)
    draw()
    clock.tick(60)
```

 **Pay attention:** Note the order in which we draw. We draw the stars first, then the spaceship and then the missiles. The things we draw first can be hidden by what we draw last. It means that we have to draw first the things that are in the back.

